



# The turn to practice in design ethics: Characteristics and future research directions for HCI research

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## HIGHLIGHTS

- Ethics in design is seen as evolving, situated, and shaped by real-world practice.
- Studies aim to support designers in navigating ethical tensions within context.
- A turn to practice in HCI helps unify ethics as an emerging research area.
- Collaborative methods help embed ethics as a dynamic design capability.
- Limited concept-building research and long-term study designs.

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## ABSTRACT

As emerging technologies continue to shape society, a growing body of work is engaging with design ethics as it unfolds in practice to better capture the complexities of ethical considerations embedded in day-to-day work. Positioned within the broader “turn to practice” in HCI, this paper brings together this body of work as an emerging research area, characterizing its motivations, conceptual positionings, methodologies, and contributions through a review conducted across a range of design disciplines and academic databases. The findings reveal a shift away from static and abstract ethical frameworks toward an understanding of ethics as an evolving, situated, and inherent aspect of design activities, one that can be cultivated and fostered collaboratively. We propose six future directions to establish common research priorities and stimulate the area’s growth. While the paper promotes cross-disciplinary dialogue, we argue that HCI research, given its cumulative experience with practice-oriented research, is well-equipped to guide this emerging research area on design ethics.

## 1. Introduction

The ethics of design in Human-Computer Interaction (HCI) has been a topic of research for decades (Shilton, 2018), and questions related to the responsibilities and moral obligations of designers appear in well-established disciplines such as participatory design (PD) (Bødker et al., 2022) and value-sensitive design (Friedman et al., 2013), sustainability (Hansson et al., 2021) as well as in work on science and technology studies (STS) and critical HCI (e.g., feminism Bardzell and Bardzell, 2011, critical race theory Williams, 2020, more-than-human Eriksson et al., 2024; Mancini, 2017, and care ethics Garrett et al., 2023; Croon, 2022). Despite this long-standing engagement with ethics, several authors have noted that much research in HCI and design has treated ethics as an *a priori* and static concept (Frauenberger et al., 2016), with less attention

paid to how ethical issues unfold in design practice (Gray and Chivukula, 2019). However, recent years have seen growing interest in considering ethics from the perspective of practice, specifically, how designers, technology developers, and the organizations they work within approach ethical issues in their everyday work. Rooted in the broader turn to practice in HCI (Bødker, 2006; Kuutti and Bannon, 2014), which emphasizes the complex and situated nature of everyday interactions, much of the research on design ethics in practice has emerged from the HCI research. This body of work highlights growing concerns about the ethical implications of design, particularly in the context of emerging technologies and their widespread impact on daily life.

Despite the number of papers focusing on empirical studies of design ethics in practice accumulating (Lindberg et al., 2021; Dindler et al., 2022; Gray and Chivukula, 2019), the research remains nascent, and

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contributions are scattered across HCI, design, and STS research venues. Therefore, we suggest it is timely to bring these efforts together, underlining their relation to the turn to practice in HCI, and reflect on their key features to help establish a more coherent and unified area of research. In this paper, we aim to position this research as an emerging area by characterizing what we describe as a “turn to practice in design ethics research”. We explore the motivations, methods, conceptual positionings, and key findings of research, and outline a range of gaps and potentials for developing future research directions through a review of contributions across HCI and design that approach ethics from the perspective of practice. Thus, this research is guided by the following research questions:

- What is the state-of-the-art in studying ethical engagements within design practices?
- What characteristics does the literature exhibit in terms of theories, concepts, and methodologies employed to explore design ethics in practice?
- Which gaps and potential future directions can be identified and grounded within HCI research?

In HCI research, design practice has drawn attention since the “turn to design” in the 1990s and the “turn to practice” in the 2010s, leading to a deeper focus on the complex, situated nature of design work (Gray et al., 2018). In particular, UX-related design work has been recognized as a distinct and valuable form of inquiry, deeply rooted in context, guided by professional expertise and ethical responsibility, and following its own standards of rigor (e.g., Søndergaard, 2020; Inie and Dalsgaard, 2020; Lee et al., 2025; Gray and Chivukula, 2019; Pink et al., 2013; Gray et al., 2022; Lim et al., 2008). Our definition and boundaries of design practice for this review draw heavily on the conceptual frameworks offered by Stolterman (2008) and by Wolf et al. (2006) as well as the articulation of Löwgren (1995) on creative design. Löwgren (1995) describes creative design work as an ongoing dialogue between identifying the problem and finding solutions, emphasizing understanding the problem just as much as developing the final product. This process often involves exploring a wide range of ideas and re-evaluating initial assumptions, making it inherently uncertain and personal to the designer. Within the scope of this paper, design practice refers to creative design practices, including UX design, as well as design work that complements or intersects with HCI, such as graphic design, industrial design, and visual communication design. Our goal is to consolidate these diverse perspectives into HCI to better address the complexities of design culture and practice within the context of technology development. This focus is reflected in the review process, where the search strategy was developed to identify papers across both HCI and design research venues. A cross-field review also demonstrated that, given its cumulative experience with practice-oriented research, the HCI community is well-positioned to lead the development of this emerging area of design ethics.

This paper offers several contributions to advancing design ethics research within HCI. First, we characterize the turn to practice in design ethics research by pinpointing shared motivations, conceptual positionings, methods, and contributions. In doing so, we aim to meaningfully bring together these diverse works under the concept of the turn to practice in HCI, and to promote a shared understanding of design ethics in practice that supports a more unified research landscape. Second, by synthesizing studies scattered across various venues and design disciplines, we seek to foster a more cohesive and dialogic research area. Third, drawing on our findings, as well as the gaps and potentials identified in the current literature, we propose several concrete directions for strengthening and expanding this area of research.

The article is organized in the following way. Section 2, Background, begins with an overview of how the concept of practice has been understood and studied across different paradigms within HCI. It then focuses on approaches from the third paradigm of HCI and the turn to practice in HCI, highlighting their relevance for research in design ethics. Section 3,

Review Process, details how the literature was collected, analyzed, and synthesized. Section 4, Results, and Section 5, Analysis, present our key findings, including: (i) the motivations driving the studies and how these informed practice-oriented research approaches; (ii) the common frameworks, research fields, and concepts in which the studies position themselves, including shared attributes of how the concept of ethics is framed across studies; (iii) the methods employed and how they address the situated, everyday nature of ethics; and (iv) the overarching research aims, findings, and contributions, with links drawn across these dimensions. Section 6, The Discussion, characterizes the turn to practice in design ethics as a research area bringing together the main insights and introducing six key directions for future work. The final section concludes by summarizing the key contributions of the review.

## 2. Background

### 2.1. The turn to practice in HCI research

The history of HCI is often described through a series of waves or paradigms, each representing a shift in focus, theoretical foundations, and methodologies (Rogers, 2012; Bødker, 2015; Harrison et al., 2007). Across these waves, the concept of “practice”—both in terms of its meaning and how it is studied—has evolved significantly. In first wave HCI, the practice in focus was a scientific and engineering-based approach to designing and evaluating human-computer interaction (Bødker, 2015). It emphasized optimizing the fit between humans and machines, mainly in work settings, using formal methods, cognitive theories, and performance metrics. Design followed analysis and aimed to improve usability by making systems more efficient, error-free, and easy to learn. It can be said that, in this view, social or cultural aspects of technology use were not central (Bødker et al., 2022; Bannon, 1991; Harrison et al., 2007).

In the second wave of HCI, the focus of practice shifted from individual users to understanding how groups use technology in real world settings (Rogers, 2012; Bødker, 2006). It emphasized social interaction, context, and collaboration within communities of practice. Theoretical frameworks such as situated action, distributed cognition, and activity theory became central (Rogers, 2012). These approaches offered new ways to understand how technology supported or disrupted the flow of everyday work. Design aimed not only for usability but also for supporting group work and addressing social and organizational issues, laying the foundation for later shifts toward understanding technology as part of broader social practices. The methods used in this wave moved away from rigid, formal testing toward more flexible and exploratory approaches such as participatory design and ethnographic fieldwork, prototyping, and contextual inquiries (Bødker, 2015; Bødker, 2006; Fallman, 2011). Although the second wave did not fully embrace the practice turn that would define later developments in HCI, it laid important groundwork. The growing interest in context, social interaction, and collective activity helped prepare the field for the more fundamental shifts that would emerge in the third wave (Kuutti and Bannon, 2014).

The third wave (Bødker, 2006; Harrison et al., 2007) extended HCI’s focus beyond the workplace into areas such as leisure, home, and public spaces. It represents a significant shift, emphasizing interaction as situated within the social and embodied complexities of everyday life (Frauenberger et al., 2016; Kim and Lim, 2023; Judge and Neustaedter, 2010). Rather than targeting clearly defined tasks or user groups, it explores how people interact with technology in diverse and often ambiguous ways. It highlights situatedness (Frauenberger et al., 2016; Kuutti and Bannon, 2014), meaning-making (Pink et al., 2013), embodiment (Frauenberger, 2020), and experience (Kuutti and Bannon, 2014)—understanding interactions as embedded in specific social, cultural, and physical contexts rather than isolated events. As an example of this move toward everyday settings, (Rogers and Marshall, 2017) propose “research in the wild” as an umbrella term for approaches that study technology in natural, real-world environments rather than controlled lab settings. They define it as an effort to understand how people use and interact with technology in everyday life, using a combination

of qualitative methods such as interviews and observations, as well as quantitative data, with a focus on how technology integrates into daily routines (Rogers and Marshall, 2017).

The concept of practice takes on a deeper and more central meaning in the third wave than in earlier paradigms. The third wave—and what is often referred to as the turn to practice—repositions practice as both the unit of analysis and the unit of design. It aims to understand the complex and situated nature of technologies embedded in seemingly mundane activities and routines (Kuutti and Bannon, 2014; Entwistle et al., 2015; Kuijer et al., 2013). Kuutti and Bannon (2014) advocate for a practice-oriented approach, encouraging research that addresses real-world challenges encountered by individuals, communities, or industries rather than centering on abstract or purely theoretical concerns. This shift draws heavily on social science theories of practice from fields such as STS, anthropology, sociology, and cultural studies (Pierce et al., 2013). For example, Reckwitz (2002) describe elements of practice (bodily and mental activities, things and their use, background knowledge, emotions, and motivation) and Shove et al. (2012) propose a framework (practice as a combination of materials, competences, and meanings) that has informed HCI approaches to studying the organization and reorganization of shared activities and routines (Kuijer et al., 2013; Lindberg et al., 2020; Entwistle et al., 2015; Kuutti and Bannon, 2014; Soden et al., 2025; Tomlinson et al., 2013). These frameworks provide empirical perspectives that consider the historical, structural, and contextual dimensions of practice, as well as interconnections with other practices (Kuutti and Bannon, 2014).

Within these theories, the concept of practice has been interpreted in various ways. It can be seen as the implicit knowledge or skills that guide human actions, or as organized sets of activities involving both people and material artifacts, or as sociomaterial arrangements closely linked to specific contexts, individuals, and interactions (Guzman, 2013). In any case, two common threads across these interpretations are: (1) practice is the setting where actions, decisions, and negotiations take place, and (2) practices are shaped by and embedded within broader technological, institutional, and infrastructural contexts (Pink et al., 2013; Entwistle et al., 2015). Waern and Back (2017) suggest understanding design knowledge as “activity”—that is, knowledge in design is not something fixed or separate from the act of designing, but rather something that emerges through the ongoing process itself. They develop a practice-oriented view of design knowledge by emphasizing that it is generated through the act of designing, where materials, ideas, concepts, and human actors co-evolve. Rogers (2004) define “the practitioner’s perspective” in research in the wild as accounts from “people who work in industry and are in the business of researching, designing, and evaluating products” (p.24). In broad terms, design practice refers to the contexts in which activities such as ideation, sketching, and prototyping are carried out, as well as where ethical decisions are made and embedded.

Amid the growing interest in ethics across HCI and design, a recent strand of work studies ethics as it unfolds in design practice to understand the complex, context-specific nature of design ethics—and the active role designers play in recognizing and engaging with ethical challenges. Upcoming sections of this paper situate this body of work within the broader turn to practice in HCI research, linking it to the epistemological foundations that characterize third wave HCI research.

## 2.2. The turn to practice in design ethics in HCI research

The question of ethics runs throughout the history of HCI and design (Shilton, 2018; Chan, 2018), and has been regularly rearticulated in response to major developments in architecture, product design, and, more recently, the widespread adoption of digital technologies—most notably, artificial intelligence. Ethics has long played a central role in traditional disciplines such as architecture, particularly in debates over how the built environment enables or impedes the good life, as seen in movements like the Bauhaus (Ramos, 2019). In product design,

ethical concerns have likewise featured prominently in ongoing discussions around sustainability, production methods, and the viability of circular economies. From an academic perspective, leading design journals have often explored the need for “discipline-specific ethics” within design (Donahue, 2004; Jonas, 2006), as well as how classical ethics frameworks such as deontology and consequentialism might provide a theoretical foundation for design ethics (Mitcham, 1995). Additionally, scholars have drawn on ethical traditions from thinkers such as Aristotle (Bousbaci and Findeli (2005), Levinas, and Derrida Steen (2012) to capture the nuanced nature of ethics in design. Despite this seemingly rich discourse, there remains skepticism about the field’s tangible progress in integrating ethical considerations into practice (Fry, 2004), prompting calls for a renewed focus on ethics in design (Chan, 2018).

In HCI, ethical considerations can arguably be traced to the very origins of the field, particularly through ongoing discussions about technology-mediated human activity; evolving from early critiques to developing generative frameworks and focusing on how ethics plays out in real-world, situated contexts (Shilton, 2018). Early scholars like Wiener (1950) contributed significantly to computer ethics, exploring issues such as privacy, surveillance, fairness, and accountability (Dindler et al., 2022; Shilton, 2018). Throughout the 1990s, within HCI, the CSCW scholarship was founded on the value of cooperation and addressed issues like empowerment and privacy. PD similarly engaged with ethical issues such as cultural norms and legitimate participation (Shilton, 2018). Fallman (2011) argues that the rise of user-centered design—strongly influenced by Scandinavian traditions—brought ethical and value-based concerns to the forefront of HCI. This included involving users early in the design process and considering their needs, values, and concerns, especially regarding power, labor, and workplace politics. According to Shilton (2018), earlier work was mainly critical, analyzing the unintended consequences of technologies and emphasizing perspectives from feminist, anti-racist, and postcolonial critiques. Later, researchers began developing generative approaches that aimed to integrate ethical considerations into the design process itself (Shilton, 2018; Steen, 2015).

In parallel, STS tradition has approached technology as shaped by social and political forces, questioning assumptions of neutrality and drawing attention to how technologies can reflect and reinforce inequalities related to gender (Bardzell and Bardzell, 2011; Van Der Velden and Mörtberg, 2012; Wagman and Parks, 2021), race (Ogbonnaya-Ogburu et al., 2020; Williams, 2020; Adamu, 2023), class (Tandon et al., 2025) and neurodiversity (Rispoli and Criado, 2024; Alrøe and Krogh, 2025). Research on the political dimensions of technology—and on the values embedded within technological systems—has become a focus across STS and HCI fields such as engineering ethics (Verbeek, 2006), reflective design (Sengers et al., 2005), and values levers (Shilton, 2013). These argue that designers’ values influence how technologies are envisioned and built, and in turn shape their broader social effects, as well as how specific practices in the design process can elevate discussion around values (Shilton, 2013).

Within third wave HCI, one of the most well-known frameworks is Value Sensitive Design (VSD), which combines theoretical, empirical, and technical methods to consider the needs of stakeholders (Shilton, 2018). VSD encourages designers to think about both direct and indirect stakeholders using tools like stakeholder analysis, scenarios, and interviews. Despite its contributions, VSD has also been critiqued for potentially favoring the values of researchers over those of participants, particularly when relying on predefined heuristics rather than more context-sensitive, participatory methods (Fallman, 2011; Shilton, 2018). Other generative methods have emerged to complement or expand upon VSD. For example, “Values at Play” and “worth-centered design” bring more artistic and creative interpretations to ethical design, focusing less on rationality and more on experiential worth (Shilton, 2018). Similarly, approaches like “critical design” and “speculative design” take a broader perspective by questioning the direction and purpose of technology itself, focusing on uncovering and addressing hidden values within the

design process, and aiming to challenge limitations and give voice to perspectives that are often overlooked. Another significant body of work addresses moral values as “inscribed” in digital artifacts—mediating actions and promoting certain values or perceptions over others (Verbeek, 2006).

Design ethics research has now been further fueled by public debates around data, privacy, and the use of AI, spurring research on issues such as dark patterns in user interfaces (Gray et al., 2023b; Lee et al., 2025; Krauß et al., 2024; Renaud and Zimmermann, 2018), and AI ethics (Khan et al., 2022). The current state of HCI and design researchers grappling with ethical practice is marked by a growing realisation of technology’s significant impact on culture and a corresponding concern for the responsibility of designers. A more recent direction in the field argues for the need to study ethics as it unfolds within design practice. Traditional, anticipatory, and formalised ethics processes, often rooted in research models and focused on avoiding risk and harm, are seen as insufficient to manage the ethical dimensions that emerge in the situated, value-driven, and participatory nature of third-paradigm HCI (Frauenberger et al., 2016). The way in which work and practice are conceptualized directly shapes how ethics is perceived and enacted. Using Gherardi’s framing (Gherardi, 2009), practice-oriented research in design ethics examines how practitioners engage with ethical questions through their work—how they think, feel, and talk about ethics—and is affected by both “outside” sources such as organizational structures or technologies (e.g., Gray et al., 2024b), or from “inside” sources like personal skills or emotions (e.g., Popova et al., 2022; Garrett et al., 2023). Since ethical questions often emerge from within practice itself, a full understanding requires a holistic view that considers the many intertwined elements shaping values and behavior (Dindler et al., 2022; Pillai et al., 2022; Gray and Chivukula, 2019; Chivukula et al., 2020).

In sum, the turn to practice in HCI and design research marked a shift from focusing on individual cognition and isolated artifacts to examining how knowledge, meaning, and design work emerge through everyday activities and social interactions. In HCI ethics research, this shift has emphasized that ethical issues are not merely theoretical concerns but are shaped by practitioners’ situated actions, negotiations, and collective sense-making within real-world work contexts.

### 2.3. Definitions

To ground our argument and clarify the terminology used in this paper, we close this Background section by offering general, working definitions. Following Frauenberger et al. (2016), we define *ethics* “as a discipline in the philosophical and systematic study of morality which is expressed in inquiries into the nature of the good life”, and *ethics frameworks* as emerging from philosophical understandings of ethics “for assessing how people should act in particular situations” (Frauenberger et al., 2016). Examples of the latter can be normative ethics and applied ethics, as well as alternative frameworks developed within HCI, such as in-action ethics (Frauenberger et al., 2016). In our analysis, we will return to which ethics frameworks are adopted or developed in design ethics research.

In studies that take a practice-oriented perspective on design ethics, we notice that the term *ethics* is used in a way that aligns with Frauenberger et al. (2016) observations in PD research, where ethics often refers to a “moral stance, or a moral attitude that underlies a particular practice.” Accordingly, when we discuss the papers reviewed for this article, we will frequently use terms that make philosophical understandings of ethics visible in practice—such as ethical sensitivity, ethical considerations, ethical deliberation, and so on.

We also clarify our use of the term *moral values*. In this paper, ethical and value-related concerns frequently appear together, reflecting how they are treated in the literature we reviewed. While we understand ethics as a philosophical system of rules or normative commitments (often tied to professional or disciplinary contexts), moral values are more personal (Gray and Boling, 2016), and enacted in everyday decisions

(Knobel and Bowker, 2011). In our reviewed corpus, values often appear as “ethical imports” (Friedman et al., 2013), referring to concepts such as privacy, consent, sustainability, participation, and related concerns. Throughout this paper, we use values as shorthand for moral values.

### 3. Review process

Our research questions and review protocol are formulated to map the breadth of the research field, suitable for the framework of the scoping review (Paré et al., 2015; Arksey and O’Malley, 2005). We look for work that reports on empirical studies involving practitioners through methods such as observation, interviews, or collaborative sessions, and work that explicitly addresses ethics as it relates to design practice. To develop a comprehensive characterization of design ethics in practice and its emerging variants, we conducted literature searches across three major databases: Web of Science (WOS), the Association for Computing Machinery (ACM) Digital Library, and the Design Research Society (DRS) Digital Library. WOS was chosen for its extensive and diverse journal coverage, including major journals within HCI, design, and STS. Our preliminary research revealed several papers on design ethics and practice within the ACM and DRS Digital Library, and it was thus also selected as a key database to include. We included full-length research articles published in English in journals and conferences indexed in these databases over the past ten years; abstracts, pictorials, and editorials were excluded. Short papers were also excluded because conferences vary widely in how they define them—ranging from work-in-progress to speculative ideas—and they typically lack the empirical depth of full papers.

#### 3.1. Identifying relevant literature: Information sources and search queries

To create a wide but focused set of search queries, we broke down our research questions into keyword categories and searched combinations of these keywords to cover the area. In addition to the keywords “design\*” and “ethic\*”, we used “design team”, “designer”, and “practitioner” to cover the context of the practice component of the research. To specify the empirical methods, we used the keywords: “interview”, “workshop”, “observation”, “qualitative”, “case study”, “fieldwork”, “codesign” and “participatory design”. We searched ethics and practice-related keywords in the “Abstract,” and method-related keywords in “All fields,” as sometimes articles do not specify their methods in the abstract. Searching with combinations of these terms generated a total of 24 search queries. The search was executed between September and November 2024.

We narrowed down the results with the publication filter: any general HCI, design, and STS-related journal and conference proceedings, as well as more scope-specific journals such as social design, visual design, and game design, were included. After completing all the queries, we checked whether a set of articles that we had handpicked and knew beforehand was included as validation of our search set.

#### 3.2. Selecting the literature: Inclusion and exclusion criteria

We conducted an initial screening of the abstracts of the papers identified during the search process. We spent a great deal of time establishing thorough inclusion and exclusion criteria because this is essential to structuring a consistent scoping review (Paré et al., 2015). To refine and finalize our inclusion criteria, article-by-article deliberations were conducted on selected papers exemplifying boundary cases. This led to the following:

- Papers were included only if their central focus was design ethics in practice; articles that merely mentioned ethics in passing were excluded.
- We focused on papers that generated insights through empirical research methods, such as interviews, observations, or case studies within design practices. Theoretical and review papers, as well as those analyzing materials such as design manifestos (Fritsch et al.,

2018), job listings (Rismani and Moon, 2023), or Reddit forums (Gray et al., 2020), were excluded, as they did not provide direct insights from design practice.

- The review intends to characterize a movement in design ethics research that we frame as a turn to practice. As such, to focus our publication pool, we omitted papers focusing on specific themes such as sustainability, vulnerable users and communities, inclusivity, and accessibility, or privacy and transparency. Our search queries caught these papers because they used the “ethic\*” keyword in their abstracts. Although these themes are related to ethics in general, including them would have expanded our scope to an unmanageable range. We thus excluded these papers to maintain our focus on design ethics in practice and to keep the number of papers manageable. We employed the same criteria for omitting papers using themes adjacent to ethics, such as responsible innovation and corporate social responsibility, as well as those offering critical perspectives on specific issues where ethics was only mentioned in passing.
- Similarly, using ethic\* as our keyword may have led to the exclusion of studies that address the social and political implications of technology—such as issues of race, gender, disability, ethnicity, refugee status, and class—without explicitly using the term ethics. We recognize that these areas are essential to understanding the contextual foundations and practical concerns of design ethics research. Even if such works may not appear in our final selection, we draw on them in the background and discussion sections to inform and situate our analysis.
- Papers on “AI ethics” appeared in large numbers in our results. In our formulation of design practice and keeping our focus on “design ethics”, we omitted these papers if they only focused on engineering design, system design, and information design. We included AI ethics and computer ethics papers only if they explicitly involved designers as research participants.
- Because educational contexts involve completely different factors and dynamics and do not fit with our main focus on design practice, we excluded papers focused on teaching ethics to design students, or curriculum studies.

After applying these criteria, we revisited our publication pool to ensure consistency and alignment with the study’s objectives. All decisions were cross-checked by the authors to maintain accuracy and rigor. Ultimately, 47 papers were selected for detailed review (see Fig. 1).

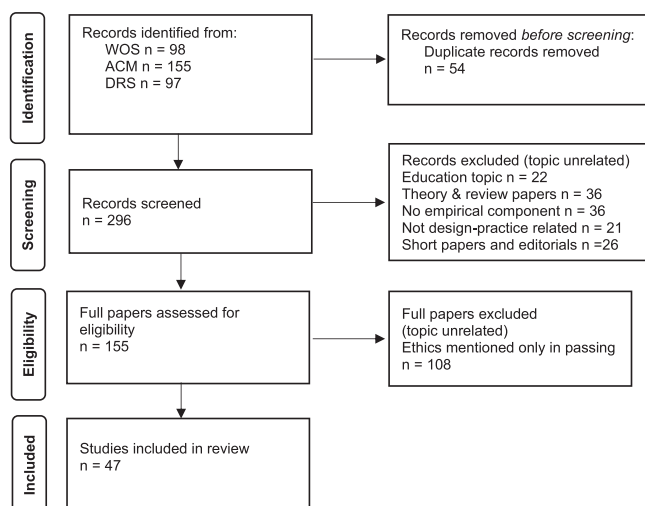


Fig. 1. PRISMA flow diagram.

### 3.3. Data charting and synthesizing of results

We adopted a descriptive-analytical method (Arksey and O'Malley, 2005) to chart the selected studies. This approach involved applying a consistent framework to examine all primary research reports and extract key details from each study. As a team, we agreed on a set of data items to extract from each paper, ensuring alignment with our main research questions. These items included both general study information and specific details about methods, themes, and findings (Arksey and O'Malley, 2005). We classified each included citation according to: (1) publication year; (2) field of design; (3) context of the research; (4) location of the research; (5) methods used; (6) research conducted by whom, with whom, and for whom; (7) main purpose of the research; (8) approaches or theories of ethics applied; (9) adjacent terms used interchangeably with ethics; (10) main study motivations; (11) main findings; and (12) the type of contribution.

Each author recorded the agreed-upon data items in a shared spreadsheet for all selected studies. When uncertainties arose—such as when a data item did not fit well within the predefined categories or when a new category seemed necessary—we met to discuss and refine our approach. To ensure consistency and accuracy, we also cross-checked each other's work by randomly reviewing data extracted by other team members.

To synthesize and summarize the charted data, we identified key themes from the literature based on our research questions and grouped them under thematic headings. For example, data items such as approaches, theories, and adjacent terms were combined under the “Conceptual Positionings” thematic heading as they are complementary to each other, while purposes, findings, and contributions were summarized together under the “Purposes and Contributions” thematic heading to highlight their relationships. We used tables to show the distribution of papers by publication year and venue, as well as to present methods, study types, main research purposes, and contributions.

### 3.4. Limitations

While we have aimed to make this review as comprehensive as possible, we acknowledge that some relevant articles may have been missed due to our keyword selection and inclusion criteria. As this is a scoping review, our primary focus has been on maintaining consistency within the dataset rather than including every possible study. Significant efforts have been put into developing and applying inclusion criteria to balance feasibility with comprehensiveness (Paré et al., 2015), as our goal is to have a manageable number of studies that provide a meaningful foundation for understanding the body of research on design ethics in practice.

As the review focuses on ethics in design practice, we have tackled the question of what constitutes “design practice” throughout the review process. We rely on the authors’ self-positioning of their research; studies are included in the review pool if the authors explicitly state that their research focuses on practice. This reliance may have led to the exclusion of research that, while relevant, did not explicitly frame itself within the context of design practice. Nevertheless, the studies we selected span a variety of practices, and we believe we have captured a level of diversity that enables us to characterize the turn to practice in design ethics.

The focus of the review on empirical research reflects the goal of capturing insights from design practices. This framing led to the exclusion of theoretical and review papers from the final selection, even when these papers were closely related to our topic. We have nonetheless tried to include these papers in the Introduction, Background, and Discussion sections to acknowledge their contributions and integrate their insights. Additionally, the search queries retrieved theory-developing papers that exemplify a set of empirical cases, such as in-action ethics (Frauenberger et al., 2016), and we included these in our selection.

Centering the review on the keyword *ethic\**—in line with our aim of examining research that engages with design ethics as it unfolds in practice—shaped the boundaries of what became visible in our dataset.

This approach may have excluded research that engages substantively with the social and political dimensions of design practice without explicitly using the term ethics. We recognize that such scholarship is essential to understanding how ethical considerations emerge in practice. To address this limitation, we draw on these bodies of work in the Background and Discussion sections to contextualize and inform our analysis.

Finally, the review focuses on papers published in the last ten years. Both benchmarked reviews (i.e., Nunes Vilaza et al., 2022) and our own search results (see Table 1) reveal a noticeable increase in publications, especially after 2020. Thus, we consider this ten-year timeframe to be sufficient to capture the growing emphasis on the turn to practice in design ethics research.

#### 4. Results

Before delving into a detailed analysis, we present an overview of key results to contextualize our review. Table 1 summarizes the distribution of publications across years and venues, highlighting a notable increase in the last five years. During the initial five years (2014–2019), nine papers were identified in our review pool. This number significantly increased in the subsequent five years (2019–2024), reaching 38 publications.

When examining publication venues, 28 papers (60%) were published in design in HCI-related venues such as DIS, CHI, and NordiCHI. Furthermore, as detailed in Table 2, 26 papers (55%) specifically position themselves within design practices in HCI, focusing on design practices such as UX design and GUI design. Many of these studies employ broad terms such as “technology design” or “technology and design practitioners” to frame their work, often emphasizing digital and interactive design practices. The participatory design (PD) field is also prominent. Some studies focus on ethics in PD more generally (Kelly, 2019; Gram-Hansen and Ryberg, 2016), while others examine the ethical considerations of designing with specific user groups, such as teenagers and children (McNally et al., 2016; Spiel et al., 2020; Malinverni and

**Table 1**  
Summary of publications by year and venue, 47 papers in total.

| Year | Instances | Venue and references                                                                                                                                                                                                                                                                                                                                                                                   |
|------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2016 | 2         | CHI (McNally et al., 2016), SIGCAS (Gram-Hansen and Ryberg, 2016)                                                                                                                                                                                                                                                                                                                                      |
| 2017 | 5         | Interacting with Computers (Malinverni and Pares, 2016; Shilton and Anderson, 2017; Frauenberger et al., 2016), CHI (Barry et al., 2017), ACM SIGDOC (Petersen, 2017)                                                                                                                                                                                                                                  |
| 2019 | 2         | CHI (Gray and Chivukula, 2019), Co-design (Kelly, 2019)                                                                                                                                                                                                                                                                                                                                                |
| 2020 | 6         | Co-design (Spiel et al., 2020), Design and Culture (Ahmed, 2020), CHI (Chivukula et al., 2020; Madaio et al., 2020), Urban Planning (Roosen et al., 2020), NordiCHI (Lindberg et al., 2020)                                                                                                                                                                                                            |
| 2021 | 7         | CHI (Chivukula et al., 2021; Wong and Nguyen, 2021), Design Studies (Rivard et al., 2021), International Journal of Technoethics (Jacobs and Wijnand IJsselsteijn, 2021), C&T (Braybrooke et al., 2021), CSCW (Lindberg et al., 2021) PIVOT (Luján Escalante et al., 2021)                                                                                                                             |
| 2022 | 11        | Design Studies (Gray et al., 2022), Games and Culture (Karlsen, 2022), International Journal of Design (Dindler et al., 2022; Nelissen and Funk, 2022), International Journal of Technoethics (Lecomte and Jean, 2022), CHI (Popova et al., 2022; Luria and Candy, 2022), NordiCHI (Pillai et al., 2022), Nordes (Lindberg et al., 2022), DRS (Luján Escalante et al., 2022; Ozkaramanli et al., 2022) |
| 2023 | 3         | DIS (Haghighi et al., 2023; Gray et al., 2023a), MobileHCI (Yoon and Jun, 2023)                                                                                                                                                                                                                                                                                                                        |
| 2024 | 11        | Design and Culture (Beattie et al., 2024), Visual Communication (Cooper, 2024), CHI (Gray et al., 2024b; Mildner et al., 2024), NordiCHI (Popova et al., 2024a), ACM Games (Millard et al., 2024), ACM J. Responsib. Comput. (Gray et al., 2024a), Proc. ACM Hum.-Comput. Interact. (Popova et al., 2024b; Lindberg et al., 2024), DRS (Chivukula and M. Gray, 2024; Tekogul and Forlano, 2024)        |

**Table 2**

Summary of design practices in the focus of the research stated by the authors.

| Design fields               | Instances | Design practices and references                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Design practices within HCI | 26        | Soma Design (Popova et al., 2024a, 2022), UX Design (Yoon and Jun, 2023; Lindberg et al., 2020, 2021; Chivukula et al., 2020; Gray and Chivukula, 2019; Pillai et al., 2022; Beattie et al., 2024; Nelissen and Funk, 2022; Wong and Nguyen, 2021), Technology Design (Gray et al., 2023a, 2024b; Haghighi et al., 2023; Chivukula et al., 2021; Chivukula and M. Gray, 2024; Gray et al., 2024a; Dindler et al., 2022; Shilton and Anderson, 2017; Frauenberger et al., 2016; Barry et al., 2017; Madaio et al., 2020; Popova et al., 2024b), Game Design (Karlsen, 2022; Millard et al., 2024), Conversational User Interface (Mildner et al., 2024) |
| Participatory design        | 5         | McNally et al. (2016); Spiel et al. (2020); Kelly (2019); Gram-Hansen and Ryberg (2016); Malinverni and Pares (2016)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Other                       | 12        | Industrial Design (Lecomte and Jean, 2022), Graphic Design (Ahmed, 2020), Communication Design (Cooper, 2024; Petersen, 2017), Urban Design (Roosen et al., 2020), Method Design (Gray et al., 2022), Various (Braybrooke et al., 2021; Jacobs and Wijnand IJsselsteijn, 2021; Rivard et al., 2021; Tekogul and Forlano, 2024; Lindberg et al., 2024, 2022)                                                                                                                                                                                                                                                                                            |
| Not specified               | 4         | Luján Escalante et al. (2021); Ozkaramanli et al. (2022); Luria and Candy (2022); Luján Escalante et al. (2022)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Pares, 2016). Contributions from other domains, such as industrial design and graphic design, are relatively infrequent. While research in the “Various” group involves participants from multiple design fields, research in the “Not Specified” group does not mention any specific design field and refers to design in general terms. When combining insights from venues and design fields, it is evident that design in HCI-related research significantly contributes to the emerging discourse on design ethics in practice. Our analysis will further explore the motivations behind this engagement, both in terms of the increasing number of studies and field-specific focus, as well as its potential implications.

Finally, while we aimed to categorize research by location, this proved challenging as most studies do not explicitly indicate the socio-geographical context of their research. It was often unclear whether the research context aligned with the location of the authors and their institutions, the origins of the study participants, or the contexts of the companies involved. Consequently, it can be said that there is a noticeable lack of contextualization regarding the geographical and cultural settings of the research, as well as the researchers’ positionality within those settings.

#### 5. Analysis

In this section, we analyze the review data in response to our research questions, focusing on the motivations behind the studies, the conceptual positionings and empirical methods employed, as well as their primary research purposes and contributions. While presenting our results, we also identify the common characteristics attributed to design ethics in these studies, highlight existing gaps, and explore the potentials emerging from the review.

##### 5.1. Study motivations

To identify the motivations for studying design ethics in practice, we will examine researchers’ perspectives on why they engage with this area, why they consider it important and timely, and what factors may be driving the increasing number of studies published each year. After considering the motivations for studying design ethics, we will explore why researchers are turning to practice in this field.

Along with acknowledging the long-established body of research examining how design methods and objects embody specific beliefs,

values, and discourses (Haghighi et al., 2023; Malinverni and Pares, 2016; Lindberg et al., 2020, 2022), several studies—particularly those from the HCI research—in our review pool highlight the growing ethical responsibility of design in the context of emerging technologies and their widespread impact on daily life (Madaio et al., 2020; Pillai et al., 2022; Lecomte and Jean, 2022; Luria and Candy, 2022; Chivukula et al., 2021; Wong and Nguyen, 2021; Frauenberger et al., 2016; Dindler et al., 2022; Haghighi et al., 2023; Ozkaramanli et al., 2022; Lindberg et al., 2020; Cooper, 2024; Ahmed, 2020). These sources express concerns about potential negative consequences of technologies, such as algorithmic bias, surveillance, privacy issues, and the amplification of societal biases; they emphasize the need for proactive ethical considerations throughout the design process to anticipate potential negative consequences, and to develop ethical frameworks that can be utilized in design practice to support this. As an example, Haghighi et al. (2023) illustrates how advancements in ubiquitous computing have influenced design ethics by developing a workshop-based method to explore ethical implications. Due to new questions raised by technological change—such as the collection and use of personal data—researchers were motivated to develop a new method for addressing emerging ethical challenges (Haghighi et al., 2023). Additionally, amid the growing ethical responsibility prompted by emerging technologies, particular attention is given to the responsibility toward users, emphasizing how the design of technology directly impacts users' behaviors and experiences (Mildner et al., 2024; Petersen, 2017). This body of research promotes user-centered approaches (Rivard et al., 2021), scrutinizes “manipulative” UX design tactics (Beattie et al., 2024), and identifies challenges faced by designers advocating for users while balancing commercial demands (Nelissen and Funk, 2022; Beattie et al., 2024).

On a different note, when it comes to design ethics in PD, some researchers reason their work based on the inherent qualities of PD itself (McNally et al., 2016; Spiel et al., 2020; Kelly, 2019; Gram-Hansen and Ryberg, 2016; Malinverni and Pares, 2016). In this body of work, there is no specific emphasis on changes brought about by technological advancements. Instead, the collaborative and empowerment-driven nature of PD is seen as inherently raising ethical considerations—such as managing diverse perspectives, ensuring equitable power distribution, and addressing the social impact of design decisions.

Nonetheless, for the majority, research on design ethics in practice is driven by a sense of urgency due to rapidly growing digital technologies to ensure that design's increasing power and influence are wielded ethically and responsibly. This motivation reflects a heightened focus on design ethics in practice to be able to produce research that can be translated into and utilized in everyday design work to uncover, examine, and embed values into design artifacts and technical elements (Chivukula et al., 2020; Lindberg et al., 2021, 2022; Jacobs and Wijnand IJsselstein, 2021; Barry et al., 2017; Spiel et al., 2020).

There is a well-recognized gap between theoretical discussions of design ethics and the practical experiences of designers (Lindberg et al., 2021, 2022; Jacobs and Wijnand IJsselstein, 2021). Academic research often struggles to translate into readily applicable tools and approaches for practitioners. Studying ethics in practice is suggested to help bridge this gap by grounding research in real-world contexts, generating insights that are relevant and actionable for designers. Sources employ ethics in practice to “resonate” (Gray et al., 2023a), “align” (Lindberg et al., 2021), and “attune” (Dindler et al., 2022) with the everyday realities of design practitioners. This approach also helps to “translate identified values into technical affordances” (Shilton and Anderson, 2017) and create “actionable interventions” (Wong and Nguyen, 2021).

## 5.2. Conceptual positionings

The studies reviewed draw on and position themselves in relation to a variety of theoretical and conceptual sources: While some build on existing ethics frameworks, some position themselves within specific research fields and paradigms, and some rely on specific concepts to

embed their research in an ongoing and accepted body of research. The purpose of this section is to clarify how studies conceptually position ethics in relation to design practice. Specifically, we distinguish between (1) explicit ethics frameworks, (2) broader research traditions that shape how ethics is approached, and (3) individual concepts that function as a lens. In this section, we explore the prevailing conceptual positions in the reviewed literature. These are non-exclusive categories; one research study can position itself in both a specific research field and a specific ethics framework at the same time, or can engage in interdisciplinary research. We end this section by identifying common conceptual attributes of design ethics observed across the reviewed studies, despite their different conceptual positionings.

### 5.2.1. Ethics frameworks

Many of the studies in our pool do not define ethics or explicitly state what is considered ethical. This is not surprising given that our review focuses on research with an empirical component—prioritizing insights from practice over pre-established definitions. Few studies (Karlsen, 2022; Lindberg et al., 2020; Chivukula et al., 2020; Gray and Chivukula, 2019) explicitly state that they avoid defining ethics because they aim to understand how designers themselves define it. However, the studies reviewed utilize several ethics frameworks to ground and understand how and when ethical decisions are made.

These include both imported and adapted frameworks such as “everyday ethics” (Gray and Chivukula, 2019; Lindberg et al., 2021), “situated ethics” (Spiel et al., 2020; Malinverni and Pares, 2016), “ethics in real-time” (Roosen et al., 2020), “pragmatist ethics” (Gray et al., 2024a, 2023a; Lindberg et al., 2020), ethical sensitivity (Boyd and Shilton, 2021), “care ethics” (Rivard et al., 2021; Braybrooke et al., 2021), “constructive ethics” (Gram-Hansen and Ryberg, 2016), and “processual ethics” (Popova et al., 2022), as well as frameworks developed within HCI scholarship “in-action ethics” (Frauenberger et al., 2016), “ethics on the ground” (Gray and Chivukula, 2019), “micro-ethics” (Spiel et al., 2020). Although these frameworks differ in various ways, they are all used to highlight and ground the situated and fluid nature of design ethics. For example, Frauenberger et al. (2016), through a case on privacy issues regarding children's data, illustrate how design ethics require real-time decisions and constant reflection. Their paper demonstrates that the initial ethical plan could not address the complexities of family dynamics, highlighting the limits of pre-defined and rule-centered approaches to handling design ethics. Their approach involves understanding the complexities, contradictions, and compromises that arise when designers grapple with ethical considerations in their daily work.

Another framework frequently encountered in the literature is “care ethics”. Studies merge situated ethics and the ethics of care (Rivard et al., 2021; Braybrooke et al., 2021; Tekogul and Forlano, 2024; Ozkaramanli et al., 2022; Petersen, 2017; Popova et al., 2024a, 2022; Spiel et al., 2020; Popova et al., 2024b), framing ethics as an “invitation to care”. This approach involves fostering a sense of empathy toward collaborators and interpreting emerging ethical issues by attending to the needs and interests of others involved in the design process. Certain sources (Popova et al., 2024a, 2022, 2024b; Luján Escalante et al., 2021) point to a move away from purely rational and theoretical frameworks toward a deeper recognition of the role of the body and emotions in shaping ethical understanding and practice.

### 5.2.2. Research fields and paradigms

Apart from drawing on specific frameworks, we also observe that some studies in our pool explicitly position themselves within particular research fields and paradigms. Unlike the previous subsection, these do not provide ethics frameworks themselves. Instead, they are utilized to explain why researchers focus on practice and context when studying design ethics.

The first point we can highlight is the third wave HCI mentioned several times as an approach that has shaped researchers' perspectives

on ethics research. As mentioned, the third wave HCI is marked by a shift toward situated perspectives that acknowledge the complex, dynamic, and value-laden nature of HCI (Malinverni and Pares, 2016; Frauenberger et al., 2016). Some sources (Frauenberger et al., 2016; Dindler et al., 2022; Malinverni and Pares, 2016; Lindberg et al., 2020; Gray and Chivukula, 2019) explicitly suggest that this shift has influenced their design ethics research, promoting a more situated, dynamic, and practice-oriented approach. Third wave HCI provides grounding in studying ethics through design practice and draws parallels between the nature of design practice and design ethics.

STS also features across the reviewed papers as shaping their research through conceptual tools, methodological approaches, and critical perspectives that inform how scholars examine responsibility in technological practice. For example, Popova et al. (2024b) draw directly on STS research strategies, integrating ideas such as the agency of artifacts and the political nature of design, and applying STS-inspired methods such as Haraway's implosion exercise and Joler and Crawford's critical mapping. Similarly, Tekogul and Forlano (2024) build on core STS concepts, notably sociotechnical imaginaries, to challenge conventional corporate futurism and instead promote relational, situated visions of design. Gray et al. (2022) position their ethics-oriented design strategies within longer-standing efforts in STS, HCI, and design to support value-sensitive and ethically grounded practice.

### 5.2.3. Other recognized concepts

A concept frequently encountered in design ethics research within the HCI field is "dark patterns" (Beattie et al., 2024; Mildner et al., 2024; Nelissen and Funk, 2022; Gray and Chivukula, 2019; Pillai et al., 2022; Millard et al., 2024). Dark patterns are often used synonymously with unethical design. This concept refers to user-interface design choices that manipulate or deceive users into making decisions they would not make if they were fully informed and aware of alternatives. It is stated that dark patterns exploit cognitive biases and well-established design principles to subtly influence user behavior, often for commercial gain. They are identified by a lack of transparency, trust, and respect for user autonomy, ultimately disregarding ethical considerations in design. Some sources (Beattie et al., 2024; Mildner et al., 2024; Nelissen and Funk, 2022) suggest that the pervasiveness of dark patterns underscores the need for greater ethical awareness and responsibility within the design community. Unlike the frameworks described earlier, dark patterns function primarily as a problem-framing concept providing a concrete vocabulary for identifying ethically problematic design decisions. A final concept encountered is "phronesis" (Barry et al., 2017), a concept from Aristotelian virtue ethics referring to practical wisdom or judgment acquired through experience in specific contexts.

Four main attributes of design ethics emerge from a thematic grouping of how reviewed studies conceptualized design ethics by drawing from these different sources:

*Ethics are inherent to design practices:* Research views ethics not as something separate that is added onto design work, but rather as an integral and inseparable part of the design process itself. Design choices inherently carry ethical weight, shaping the values and norms embedded in technologies and affecting how people interact with the world. This recognizes that design is a form of "doing ethics" (Dindler et al., 2022; Frauenberger et al., 2016; Popova et al., 2024a).

*Ethics are contextually dependent:* The sources strongly emphasize the situated nature of design ethics. This means that ethical considerations in design are not absolute or universal; rather, they are shaped by the specific context in which design occurs (Frauenberger et al., 2016; Haghighi et al., 2023; Rivard et al., 2021). This is a dynamic and context-sensitive process that is influenced by various factors, including the designer's values, organizational culture, stakeholder needs, and the specifics of the design project (Dindler et al., 2022; Lindberg et al., 2021; Popova et al., 2024b).

*Ethics are not static but evolving:* The review highlights that ethical considerations are dynamic, evolving over time. This evolving nature

of ethics in design practice challenges rigid, rule-based approaches and points to the need to adopt a more responsive, flexible understanding, as well as emphasizes the temporal aspect of ethical encounters (Frauenberger et al., 2016; Lindberg et al., 2020).

*Ethics can be cultivated:* Several sources argue that ethical practices can be developed and nurtured over time. Ethical practices, therefore, can be negotiated and co-constructed within the context of each project, as well as within wider design cultures (Ozkaramanli et al., 2022; Lindberg et al., 2021; Luján Escalante et al., 2022).

### 5.3. Methods used

This section will examine different methods employed in the literature and the methods-related findings of our review. Because conducting empirical research was a key filtering criterion for this review, the methods observed align with this focus. While categorizing the methods used, we will also examine how the research utilizes these methods to address the hands-on, everyday quality of ethics and how they capture the implicit and often tacit aspects of ethical reasoning that may arise in the design process. In addition to these observational methods, we will also consider research that uses interventionist methods aimed at cultivating and operationalizing ethics within the design process.

Table 3 shows that the interview method is the most commonly used approach in the reviewed studies. If we include mixed-methods studies with an interview component, around 51% of the research incorporates interviews as a data-collection method. Researchers use interviews to gain insights into practitioners' perspectives on ethics in their work and organizations, to identify ethical challenges and opportunities within specific design contexts (Dindler et al., 2022; Rivard et al., 2021; Lindberg et al., 2021; Jacobs and Wijnand IJsselsteijn, 2021; Chivukula et al., 2020; Beattie et al., 2024; Millard et al., 2024), and to explore specific dimensions of ethical decision-making, such as the affective dimension (Popova et al., 2024b), identity dimension (Chivukula et al., 2021), and business models (Karlsen, 2022). Among the interview studies, some incorporate designerly approaches into the interview process, such as speculative enactments (Nelissen and Funk, 2022), probe studies (Luria and Candy, 2022; Popova et al., 2022), and visual trigger materials (Lindberg et al., 2021) to stimulate conversation,

**Table 3**  
Summary of methods employed.

| Method        | Instances | References                                                                                                                                                                                                                                                                                                                                                                  |
|---------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interviews    | 16        | Rivard et al. (2021); Karlsen (2022); Jacobs and Wijnand IJsselsteijn (2021); Lindberg et al. (2021); Chivukula et al. (2020); Petersen (2017); Dindler et al. (2022); Chivukula et al. (2021); Mildner et al. (2024); Kelly (2019); Gray et al. (2022); Popova et al. (2024b); Beattie et al. (2024); Nelissen and Funk (2022); Millard et al. (2024); Gray et al. (2023a) |
| Workshops     | 11        | Lindberg et al. (2020, 2022, 2024); Popova et al. (2024a); Chivukula and M. Gray (2024); Gray et al. (2024b); Roosen et al. (2020); Gram-Hansen and Ryberg (2016); Pillai et al. (2022); Gray et al. (2024a); Haghighi et al. (2023)                                                                                                                                        |
| Case studies  | 6         | Lecomte and Jean (2022); Barry et al. (2017); Shilton and Anderson (2017); Spiel et al. (2020); Luján Escalante et al. (2022); Frauenberger et al. (2016)                                                                                                                                                                                                                   |
| Mixed-methods | 8         | Interview + Observations (Gray and Chivukula, 2019; Tekogul and Forlano, 2024; Yoon and Jun, 2023)<br>Interview + Survey (Ahmed, 2020; McNally et al., 2016)<br>Interview + Probes Study (Luria and Candy, 2022; Popova et al., 2022)<br>Interview + Workshop (Madaio et al., 2020)                                                                                         |
| Other         | 6         | Autoethnography (Malinverni and Pares, 2016), Conversation (Ozkaramanli et al., 2022), Dancing (Luján Escalante et al., 2021), Design Sprint (Braybrooke et al., 2021), Design Activity (Wong and Nguyen, 2021; Cooper, 2024)                                                                                                                                               |

address difficult topics, and encourage future-oriented thinking during interviews.

There are six case studies within the review pool. Three research papers present multiple case studies together to offer a broader perspective on the design ethics (Frauenberger et al., 2016; Spiel et al., 2020; Shilton and Anderson, 2017), while single case studies focus on a specific company (Lecomte and Jean, 2022), or a specific product development process (Barry et al., 2017; Luján Escalante et al., 2022). For example, Shilton and Anderson (2017) uses three ethnographic case studies to illustrate the different ways of enacting the concept of values levers (Shilton, 2013), which are development practices that facilitate conversations about values and build consensus around their importance in design. Their findings highlight that (1) having an ethical design advocate role in a design team helps normalize conversations around ethical concerns, (2) understanding a team’s existing values and assumptions helps guide meaningful discussions, and (3) open conversations and consensus-building prove more effective than enforcing external rules.

The literature also contains a variety of less common methods that derive from design and arts research, such as designing activities (Cooper, 2024; Wong and Nguyen, 2021), using dancing as a method (Luján Escalante et al., 2021), employing design sprints (Braybrooke et al., 2021), and the use of probes and fictional scenarios in interviews (Luria and Candy, 2022; Nelissen and Funk, 2022). As shown in Table 4, 36% (17 studies) employed interventional methods such as co-creation workshops and soma design workshops. Co-creation and co-design workshops often involve hands-on activities in which participants collaboratively develop tangible outputs, such as ethical toolkits (Lindberg et al., 2020), action plans (Gray et al., 2024b; Chivukula and M. Gray, 2024), checklists (Madaio et al., 2020), and design artifacts (Lindberg et al., 2022; Haghighi et al., 2023) to bring together researchers and practitioners to explore ethical issues in design collaboratively. For example, Gray et al. (2024b) held co-creation workshops with design and technology practitioners to help them create personalized ethics-related action plans. The workshops aimed to empower participants to tackle ethical challenges using their design expertise. They identified key roles participants took on (i.e., the advocate, the operationalizer, the reformer) and three strategies they used to define their ethical problem space (i.e., refining, expanding, diverging). Another study used body-based workshops to explore the concept of “felt ethics”, centering around the idea that ethics can be experienced as a bodily sensation (Popova et al., 2024a). The workshop methodology is grounded in a soma design approach, which uses the body as a site for exploration and reflection on design practice to probe themes like discomfort, boundaries, control, and negotiation.

Finally, we observe a greater prevalence of cross-sectional methods compared to longitudinal ones (see Table 5). Cross-sectional research involves collecting data from each participant at a single point in time, providing a snapshot of a situation—for example, through interviews. Repeated cross-sectional research collects data at multiple time points, typically from different samples, or occasionally from the same participants in different settings over short durations, such as in a series of workshops. Longitudinal research involves collecting data from the same participants across multiple time points over an extended period, with the goal of studying changes and processes as they unfold over

**Table 4**  
Distribution of method types.

| Interventional | Observational |
|----------------|---------------|
| 36%            | 64%           |

**Table 5**  
Distribution of study designs.

| Cross-sectional | Repeated cross-sectional | Longitudinal |
|-----------------|--------------------------|--------------|
| 62%             | 30%                      | 8%           |

time—for instance, through participant observation (Karapanos et al., 2021). Current research mainly relies on cross-sectional methods, such as interviews, and repeated cross-sectional methods, such as conducting a series of workshops. However, we note the limited number of longitudinal studies that capture the temporal aspects of ethics. We see this as a significant gap in the research, especially given the emphasis on the situated and evolving nature of design ethics.

5.4. Purposes and contributions

In this section, we will identify the primary research purposes, draw parallels between these purposes and the contributions of the studies, and provide examples of the research findings. We coded the main purposes into four nonexclusive thematic categories, recognizing that a single study can fall under multiple categories simultaneously, which are: understand (how design ethics unfolds in everyday practice), support (practitioners in addressing ethical situations), operationalize (ethical considerations within design practice and organizations), and cultivate (ethical sensitivities and cultures of design ethics). Table 6 shows the distribution of research numbers according to their primary purposes; highlighting the emerging nature of the field, as the majority of research is directed towards understanding the dynamics of ethics in practice.

For categorizing contributions, we follow Wobbrocks’ article (Wobbrock and Kientz, 2016), which identifies seven types of contributions in HCI research: empirical, methodological, survey, artifact, dataset, theoretical, and opinion. Given that one of our primary inclusion criteria is that studies must include an empirical component, all the works reviewed provide empirical contributions. Additionally, we encountered (i) methodological contributions, including the development of methods to support ethical practice, (ii) theoretical contributions derived from empirical insights (see Table 7), and (iii) one artifact contribution where a website was designed to assist practitioners in finding relevant ethical decision-making tools tailored to their specific contexts (Gray et al., 2023a). Now, we delve into each purpose category in detail,

**Table 6**  
Categories of main research purposes.

| Purpose                                                                   | Instances | References                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Understand (how design ethics unfolds in everyday practice)               | 29        | Ahmed (2020); Petersen (2017); Gray et al. (2022); Rivard et al. (2021); Karlsen (2022); Dindler et al. (2022); McNally et al. (2016); Kelly (2019); Lindberg et al. (2021); Beattie et al. (2024); Chivukula et al. (2021); Yoon and Jun (2023); Popova et al. (2024b); Ozkaramanli et al. (2022); Millard et al. (2024); Gray et al. (2024a); Popova et al. (2022); Lindberg et al. (2024); Chivukula and M. Gray (2024); Gray and Chivukula (2019); Gray et al. (2024b); Madaio et al. (2020); Nelissen and Funk (2022); Shilton and Anderson (2017); Mildner et al. (2024); Gray et al. (2023a); Chivukula et al. (2020); Pillai et al. (2022); Luria and Candy (2022) |
| Support (practitioners in addressing ethical situations)                  | 17        | Luján Escalante et al. (2021); Popova et al. (2024a); Roosen et al. (2020); Malinverni and Pares (2016); Wong and Nguyen (2021); Braybrooke et al. (2021); Cooper (2024); Lecomte and Jean (2022); Haghighi et al. (2023); Spiel et al. (2020); Nelissen and Funk (2022); Shilton and Anderson (2017); Mildner et al. (2024); Gray et al. (2023a); Chivukula et al. (2020); Pillai et al. (2022); Luria and Candy (2022)                                                                                                                                                                                                                                                   |
| Operationalize (ethical considerations within practice and organizations) | 10        | Lindberg et al. (2022); Luján Escalante et al. (2022); Jacobs and Wijnand IJsselstein (2021); Gram-Hansen and Ryberg (2016); Barry et al. (2017); Frauenberger et al. (2016); Spiel et al. (2020); Chivukula and M. Gray (2024); Gray and Chivukula (2019); Madaio et al. (2020)                                                                                                                                                                                                                                                                                                                                                                                           |
| Cultivate (ethical sensitivities and cultures of design ethics)           | 3         | Tekogul and Forlano (2024); Lindberg et al. (2020, 2022)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**Table 7**  
Types of main research contributions.

| Contribution                     | Instances | References                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Empirical                        | 26        | Tekogul and Forlano (2024); Rivard et al. (2021); Karlson (2022); Shilton and Anderson (2017); Gray et al. (2024b); McNally et al. (2016); Madaio et al. (2020); Lindberg et al. (2022); Ozkaramanli et al. (2022); Chivukula et al. (2020); Petersen (2017); Dindler et al. (2022); Chivukula and M. Gray (2024); Yoon and Jun (2023); Pillai et al. (2022); Chivukula et al. (2021); Beattie et al. (2024); Ahmed (2020); Millard et al. (2024); Kelly (2019); Gray et al. (2022); Popova et al. (2022, 2024b); Lindberg et al. (2024, 2021); Gray et al. (2024a) |
| Empirical<br>+<br>Methodological | 11        | Haghighi et al. (2023); Malinverni and Pares (2016); Braybrooke et al. (2021); Lindberg et al. (2020); Luján Escalante et al. (2021); Cooper (2024); Lecomte and Jean (2022); Luria and Candy (2022); Nelissen and Funk (2022); Wong and Nguyen (2021); Popova et al. (2024a)                                                                                                                                                                                                                                                                                       |
| Empirical<br>+<br>Theoretical    | 9         | Gram-Hansen and Ryberg (2016); Mildner et al. (2024); Jacobs and Wijnand IJsselsteijn (2021); Gray and Chivukula (2019); Luján Escalante et al. (2022); Frauenberger et al. (2016); Barry et al. (2017); Spiel et al. (2020); Roosen et al. (2020)                                                                                                                                                                                                                                                                                                                  |
| Empirical<br>+ Artifact          | 1         | Gray et al. (2023a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

aligning them with their corresponding contribution categories and key findings. This approach ensures a clear connection between the purpose, its contributions, and the insights gained.

#### 5.4.1. Understand

This category contains research that aims to understand how designers engage with ethics in their practice and how different factors affect ethical positioning within their practice. The contribution type in this form of inquiry is mainly empirical.

These studies aim to gather insights into how designers perceive and engage with ethics in their work, as well as the concepts, values, and challenges shaping their ethical decision-making process (Lindberg et al., 2021; Dindler et al., 2022; Chivukula and M. Gray, 2024; Chivukula et al., 2020). By examining design practices, these studies identify factors influencing ethical outcomes, such as the role of identity and belief in ethical positioning (Chivukula et al., 2021), the affective dimensions of ethics (Popova et al., 2024b, 2022), team dynamics around ethics (Shilton and Anderson, 2017), and the vocabularies practitioners use to negotiate ethical concepts (Pillai et al., 2022; Gray et al., 2024a). Additionally, they explore the ethics of participation (Kelly, 2019; McNally et al., 2016; Spiel et al., 2020), as well as ethics in specific technologies, including artificial intelligence (Yoon and Jun, 2023; Madaio et al., 2020), mixed reality (Millard et al., 2024), and conversational interfaces (Mildner et al., 2024) and issues such as privacy (Beattie et al., 2024; Nelissen and Funk, 2022), and monetization (Karlson, 2022).

One of the prominent findings to highlight in this area is that practitioners often perceive ethics as a vague and challenging concept to define. As previously noted, some researchers choose not to define ethics in their studies, aiming instead to explore how designers themselves define it; however, many practitioners struggle to articulate individual definitions, often relying on adjacent terms like sustainability or privacy to describe ethical considerations (Beattie et al., 2024; Dindler et al., 2022). Beattie et al. (2024) describe ethics as a “floating signifier”, a term lacking fixed meaning and easily attached to various domains or concepts, illustrating the inherent ambiguity of the concept in practice.

#### 5.4.2. Support

Research in this category aims to support practitioners in addressing ethical dilemmas, implications, and decisions in their work by making

ethical considerations more accessible and actionable in everyday design practice. The contributions in this area are primarily methodological and empirical (Wobbrock and Kientz, 2016).

This type of research can develop methods such as visual mapping methods (Cooper, 2024; Wong and Nguyen, 2021; Roosen et al., 2020), body-based methods (Luján Escalante et al., 2022; Popova et al., 2024a), or speculative design methods (Haghighi et al., 2023; Luria and Candy, 2022; Nelissen and Funk, 2022). They can build collaborative and code-sign spaces (Braybrooke et al., 2021; Lindberg et al., 2022) to tackle the complex and context-dependent nature of ethics in design practice. For example, “design timescapes” (Cooper, 2024) is a method to help designers visualize the past, present, and future of their work, linking it to societal changes for better-informed decisions. Another example is “timelines” (Wong and Nguyen, 2021); a method that uses world-building to explore the ethical implications of technologies or artifacts through narratives. Combining design fiction, scenario planning, and value-sensitive design, it claims to offer a flexible way to advocate for values and examine ethics.

These studies share a common finding: while a wide range of tools and methods exist to support designers in addressing ethical concerns, they are often underused in practice. Many ethical design tools and methods are perceived as overly abstract or disconnected from the realities of daily design work. Practitioners often find that academic frameworks for ethical design fail to translate into actionable steps that can be integrated into their workflows (Gray et al., 2024b; Lindberg et al., 2020, 2021; Rivard et al., 2021; Madaio et al., 2020). Research in this category specifies that support mechanisms with insights from actual lived experiences should be built to bridge this gap. For example, a prominent term, “practice resonance” (Gray et al., 2023a), refers to the degree to which a design method aligns with the practical needs, values, and work contexts of practitioners, thereby enabling them to effectively engage with and apply the method in their everyday work. This term highlights the need to base design-ethics research on practitioners’ lived experiences, and it stresses that any method meant to help practitioners “successfully navigate, perturb, and potentially resolve” (Gray et al., 2024b) ethical complexity should be grounded in those experiences.

#### 5.4.3. Operationalize

Research in this category provides ways to understand design ethics in practice, making ethical considerations observable and enabling their operationalization within design work and organizational structures. While research in the Support category develops tools and methods intended for use at specific moments to help practitioners address ethical dilemmas, studies in the Operationalize category aim to translate ethical considerations into actions throughout the entire work process. Support tools are often designed for individual or team use, whereas research in the Operationalize category seeks to influence practices both within and across organizations (Gram-Hansen and Ryberg, 2016; Barry et al., 2017; Gray et al., 2024b; Chivukula and M. Gray, 2024; Gray and Chivukula, 2019; Spiel et al., 2020). The main contributions of this type of research are theoretical and empirical.

As an example, the ethical mediation framework and the concept of ethical design complexity developed by Gray, Chivukula, and their collaborators are prominent contributions in which they “position ethics as an important mediator of design complexity” (Gray and Chivukula, 2019). Through a series of empirical studies, they construct the concept of ethical design complexity, defined as “complex and choreographed arrangements of ethical considerations” that practitioners must navigate. This complexity emerges from the interplay of individual values and beliefs, organizational practices and cultures, and applied ethical frameworks. In essence, in this context, ethical mediation refers to the ongoing process of navigating this complexity by recognizing multifaceted ethical challenges, engaging in reflective dialogue, and taking actions aligned with responsible and impactful design practices (Gray et al., 2024b; Chivukula et al., 2020; Gray and Chivukula, 2019).

By elaborating on how complexity is structured and situating ethical mediation as the mechanism to navigate it, their work provides a theoretical framework to operationalize design ethics.

Another important example is micro-ethics, as outlined by Spiel et al. (2020) through a series of participatory design processes in which designers collaborate closely with participants, often in sensitive settings. They position ethics as situated and processual, emphasizing care, empathy, and responsibility in interactions. Micro-ethics calls for designers to be more attentive and responsive to the ethical nuances of their interactions and to cultivate an ethical awareness that permeates the entire design process. It operationalizes ethics by making the often unspoken or overlooked ethical aspects of the process more visible, focusing on the small, everyday decisions and actions that unfold throughout the design process.

#### 5.4.4. Cultivate

Cultivation refers to the ongoing, deliberate shaping of thinking and doing that guides practitioners in navigating ethical questions in their everyday work. Research on design ethics in practice treats ethics as situated and evolving; accordingly, research suggests ethical cultures (Lindberg et al., 2020), ethical sensitivities (Boyd and Shilton, 2021), and shared ethical understandings (Frauenberger et al., 2016) can be intentionally cultivated within design practice. This type of research contributes both methodologically and empirically.

For example, following Frauenberger et al.'s in-action ethics (Frauenberger et al., 2016), a collection of works by Lindberg and colleagues (Lindberg et al., 2020, 2022, 2021) engages with the concept of ethos building and expands it towards creating an ethical foundation and ethical culture. Ethos building involves a conscious and sustained effort to create a shared understanding of moral values, integrate those values into the design process, and foster a culture that supports ethical deliberation. Instead of addressing ethical considerations as they arise, a cultivation approach proactively engages in ethos building. Lindberg et al. emphasize ethos building as (1) a process that involves ongoing reflection, dialogue, and negotiation, (2) a collaborative effort among designers, stakeholders, and the broader community, and (3) a situated practice in which the specific values and commitments that constitute an ethos are shaped by context. Lindberg et al. conduct interactive workshops, often using co-design or co-creation methods, to bring practitioners together to explore the meaning of design ethics and related challenges. In these workshops, participants discuss and co-design potential solutions for addressing ethical tensions, identify obstacles to ethically responsible design, suggest strategies to overcome them, and examine how designers can cultivate their own ethical practices (Lindberg et al., 2020, 2022).

Another approach to cultivation comes from the ethical sensitivity (ES) framework (Boyd and Shilton, 2021). It provides a practical framework for examining how people engage with ethical issues at work by breaking ethical engagement into three observable components: recognizing that a situation has an ethical dimension, interpreting or making sense of the situation's specific details, and deciding how to act. Boyd and Shilton (2021) argue that the framework highlights the processes that support and shape ethical decision-making; thus, it can help cultivate ethical sensitivity by helping researchers and practitioners identify the "cues" that prompt ethical awareness, as well as additional, previously unrecognized cues.

## 6. Discussion

In recent years, there has been a noticeable rise in practice-oriented research on design ethics within HCI and design fields, as reflected in the numbers presented in our research (see Table 1). The studies in our review appear to align in meaningful ways, forming a relatively coherent research area. However, despite their prominence, these works remain dispersed across various venues and design disciplines. The turn to practice in HCI provides a useful framing that allows us, through

the notion of practice, to consolidate this body of work as an emerging research area.

This paper synthesizes existing literature on ethics from a practice-oriented perspective across design fields. In addition to ACM DL, we deliberately included both the WOS and DRS libraries, observing traces of ethics in practice across other design disciplines (see Tables 1 and 2) to encourage cross-pollination and a richer understanding. Nonetheless, it is clear that there are considerably more studies in our pool from HCI research. There may be several reasons for this. (1) As noted, much of the research is motivated by the increasing influence of emerging technologies (Pillai et al., 2022; Chivukula et al., 2021; Dindler et al., 2022; Lindberg et al., 2020). Although these technologies impact all areas of work, the focus on emerging digital technologies makes practice-oriented ethics in HCI research more dominant, as designers are often seen as the creators of these technologies. (2) The shift toward practice in HCI studies, as previously discussed, has further supported and shaped design ethics research in this domain. For these reasons, we believe that HCI research can guide design ethics in practice research across other fields. In our findings, we synthesize a characterization of the turn to practice in design ethics research in terms of its approach to the concept of ethics, its shared motivations, research purposes, and methods. In this discussion section, first, we will present a summary of this characterization and then suggest future directions for advancing this field.

### 6.1. Characteristics of design ethics research in practice

To sum up, the studies reviewed share a common motivation to turn to practice to study ethics, namely to address the growing urgency of practicing design ethics amid increasingly complex professional realities and to respond to a persistent gap between theoretical discourse and practitioners' lived experiences (Chivukula et al., 2020; Lindberg et al., 2021; Jacobs and Wijnand IJsselstein, 2021; Spiel et al., 2020). Their specific research purposes to address these main motivations converge on understanding how ethics unfolds in everyday practice, supporting designers as they navigate challenges, and developing tools that operationalize ethical considerations, while broader systemic efforts to embed ethics within organizational cultures remain comparatively limited (Luján Escalante et al., 2021; Cooper, 2024; Lindberg et al., 2022). Research approaches the concept of ethics not as a fixed set of principles but as an inherent, contextual, and continually evolving aspect of design work—something that can be cultivated through reflective and collaborative engagement (Frauenberger et al., 2016; Gray and Chivukula, 2019; Lindberg et al., 2020). Methodologically, this body of work relies on both observational approaches to examine ethical dynamics and interventional methods to scaffold ethical practice, reflecting a broader shift from abstract ethical principles toward the situated and design-relevant realities of practice (Karlsen, 2022; Popova et al., 2024a; Gram-Hansen and Ryberg, 2016).

All in all, with some research already explicitly positioning itself within the turn to practice in HCI (Frauenberger et al., 2016; Dindler et al., 2022; Malinverni and Pares, 2016; Lindberg et al., 2020; Gray and Chivukula, 2019; Gray et al., 2018), practice-oriented design ethics research shares many of the same characteristics as this broader shift. Following the turn to practice in HCI, this area of design ethics research (1) represents a move from abstract, isolated perspectives to an embedded, practitioner-centered view of real-world work, (2) emphasizes understanding situated practices, acknowledging that interaction and ethics are not standalone concerns but are, embedded in the complex realities of design and technology work, and (3) aims to bridge the research-practice gap by engaging directly with practitioners, valuing their knowledge, and addressing practical needs through context-sensitive methods like ethnography, co-design, and participatory inquiry. Thus, what we characterize as the turn to practice in design ethics exemplifies this shift by studying how ethical concerns emerge

and aretating the importance of relevance, situatedness, and practitioner engagement in both ethical reflection and research methodology.

## 6.2. Future directions

In this section, we will identify key gaps and potentials that emerged from our findings and propose six future directions that need to be addressed to advance research on design ethics in practice. Each proposed direction will be anchored in existing HCI literature, providing a deeper exploration of these gaps and outlining how they can evolve and contribute to future research in this area.

### 6.2.1. Developing concepts through empirical research

As illustrated in Tables 6 and 7, the majority of research (60%) in our pool aims to explore the current landscape of ethics in practice. Research on practice-oriented ethics is growing; however, with notable exceptions (e.g., Frauenberger et al., 2016; Gray and Chivukula, 2019), concept-building through empirical research remains limited. We believe it is timely to advocate for more context-aware, concept-building research grounded in lived experience—research that contributes to a shared conceptual foundation for practice-oriented design ethics by reflecting on common concerns and interests—(1) to bridge the theory-practice gap in design ethics, as one of the primary motivations for researchers to turn to practice (Lindberg et al., 2021, 2022; Jacobs and Wijnand IJsselstein, 2021), and (2) to ensure that a rapidly growing field does not lose direction without clear purposes or standards to evaluate its value and contribution (Rogers, 2012; Velloso and Hornbæk, 2025). Our guiding question for this future direction is: What would concepts, frameworks, and theories in design ethics look like if they were meant to resonate with, and emerge from, everyday work practice?

The theory-practice gap has long been a point of discussion in broader HCI research (Rogers, 2012; Velloso and Hornbæk, 2025), with numerous contributions aimed at addressing this divide—contributions that design ethics research can benefit from. We suggest building on these existing discussions in HCI to initiate a more focused conversation within ethics-related research. This gap arises from the abstract nature of theory, the fast-paced, pragmatic demands of design work, limited communication between researchers and practitioners, and the complexity or inaccessibility of certain theoretical frameworks (Rogers, 2012; Dalsgaard and Dindler, 2014). Additionally, it has been argued that HCI's interdisciplinary nature, coupled with differing evaluation criteria in academia and practice, further exacerbates this gap (Rogers, 2012; Velloso and Hornbæk, 2025). To address these challenges, researchers have proposed various bridging strategies.

One common strategy, as Rogers points out (Rogers, 2012), is to “import” theories from other fields to build correlations and make sense of design situations. An example from our review is that drawing on Activity Theory, Gray et al. develop the framework of ethical mediation (Gray and Chivukula, 2019), which consists of factors that shape a designer's ethical awareness and decision-making in UX practice. Ethical mediation helps explain the ethical complexity of design by demonstrating how various influences continuously shape ethical engagement. Another strategy Rogers suggests is the development of “wild theories” (Rogers, 2012)—theories that emerge directly from messy, real-world contexts, addressing the interdependencies between design, technology, and behavior. These theories in HCI serve multiple purposes—not only to predict or prescribe, but also to generate ideas and sensitize designers to critical issues (Rogers, 2012; Velloso and Hornbæk, 2025). For example, Oulasvirta and Hornbæk (2022) postulate that theories inform design practices through counterfactual thinking, enabling designers to speculate about the potential consequences of different design choices. This approach, when applied to concept development through empirical, practice-based research, can help bridge the gap between theory and practice in design ethics research.

In response to the need for more practically resonate approaches, many recent HCI efforts have focused on creating intermediary-level

concepts instead of developing formal, scientific theories (Rogers, 2004; Velloso and Hornbæk, 2025). For example, Höök and Löwgren (2012) introduced strong concepts, which are abstracted from specific design cases and are meant to enrich designers' repertoires. Though they may be supported by theory, they primarily emerge from practice. Another proposal comes from Dalsgaard and Dindler (2014), who introduce bridging concepts—a form of intermediary knowledge that explicitly connects abstract theory with practical design. Unlike strong concepts, which primarily emerge from recurring design patterns, bridging concepts are grounded in both theoretical frameworks and real-world design examples. Their primary purpose is to facilitate a two-way exchange between theory and practice, holding both accountable, and helping to surface new design opportunities and theoretical insights. Such efforts emerging from HCI research can help guide design ethics research by fostering shared concepts and contributing to a more unified research landscape, while also helping to bridge the gap between theory and practice.

### 6.2.2. Conducting longitudinal studies

The sources we reviewed showcase a wide range of methodologies used to investigate design ethics in practice, with interviews being the dominant form of inquiry. Many studies utilize designerly methods and practices to address ethical challenges, such as design tools and techniques to explore ethical dilemmas, develop strategies for ethical intervention, and facilitate discussions about values and responsibilities in design. However, one clear gap we identified is the need for long-term studies; most research in the field relies on cross-sectional study designs. Interviews, while effective for capturing specific moments in time, often fall short of fully capturing situated practices, and are less effective in exploring processes and changes over time (Jarrahi et al., 2021; Chapman, 2001). Since the ethics in practice approach views ethics as dynamic rather than static, long-term studies are essential to capture the evolving and temporal nature of ethics in design practice. Longitudinal studies can allow researchers to observe how ethical practices develop, how designers' ethical awareness grows, and how external factors influence ethical decisions.

Longitudinal research in HCI involves collecting data from the same participants at multiple points in time, allowing researchers to study how user experiences, behaviors, and perceptions evolve (Karapanos et al., 2021). While valuable, long-term studies present several challenges. They are time-consuming and costly, requiring sustained participant engagement, careful planning, advanced training, and often facing methodological issues such as inconsistent data, small sample sizes, and complex analysis, which can make achieving valid results difficult (Harbich and Hassenzahl, 2016; Karapanos et al., 2021). Despite a notable shift in HCI toward studying temporality more explicitly as a central aspect of research (Wiberg and Stolterman, 2021), similar to design ethics research, it is stated that there remains a need for more longitudinal studies in HCI in general, partly due to a lack of shared understanding and clear definitions within the HCI research (Bargas-Avila and Hornbæk, 2011; Harbich and Hassenzahl, 2016; Karapanos et al., 2021). To address this, Kjærup et al. (2021) note that researchers have been developing taxonomies to categorize types of longitudinal research questions and designs. Conferences like CHI have hosted workshops and discussions to build consensus on best practices, and literature reviews are examining how previous work has approached longitudinal study design (e.g., Jain et al., 2010; Karapanos et al., 2012). These efforts to create a common understanding can also guide ethics research.

Additionally, we believe sharing behind-the-scenes insights into design ethics could be valuable. This is an evolving area where researchers often deal with sensitive subjects and contested issues. Many researchers collaborate with design practitioners in commercial settings, where accessibility barriers, such as time constraints, economic pressures, and confidentiality concerns, are prevalent. By offering self-reflective experiences, strategies for conducting research in commercial settings, working with practices rich in implicit knowledge, and addressing potentially

triggering issues, researchers can help guide others in conducting more long-term studies.

### 6.2.3. Contextualizing design ethics research

Design ethics research emphasizes the embedded nature of ethics, highlighting its situatedness within specific design practices, designer identities, organizational contexts, and broader socio-geographical conditions. These wider conditions shape the values and principles guiding both organizations and practitioners (Madaio et al., 2020; Tekogul and Forlano, 2024; Lindberg et al., 2022; Ozkaramanli et al., 2022). While ethics is widely recognized as a sociocultural construct that varies across cultures and geographies, there is a noticeable lack of research that fully contextualizes and localizes findings. Most research does not specify its location or relate its findings to the socio-geographical context in which the authors, participants, and stakeholders are situated. A notable exception is the work of Beattie et al. (2024), who contextualizes the relationship between design practitioners and ethical decision-making within the broader framework of policies—or the lack thereof—in New Zealand. This underscores the need to reflect on how research findings align with the local nuances of where they are conducted, ensuring resonance with specific cultural and geographical contexts.

Understanding social context has become a central, though contested, concern in HCI (Räsänen and Nyce, 2006; Rogers, 2012; Joshi et al., 2024). Initially, context was often equated with “immediate” tasks and interactions, often reduced to observable actions, overlooking broader socio-cultural influences (Räsänen and Nyce, 2006). This shifted with the adoption of ethnographic methods in the 1980s within STS, which emphasized the complexities of users’ everyday work environments and the social dynamics shaping technology use (Joshi et al., 2024). Influential work, such as Lucy Suchman’s Situated Action theory, highlighted the improvisational and situated nature of interaction, challenging static views of context (Rogers, 2012). STS also provided other methodological tools—socio-material analysis (Popova et al., 2024b), and approaches for tracing distributed agency (Frauenberger, 2020)—that help researchers examine how responsibilities, values, and constraints emerge within context. More recently, a cultural turn in HCI has expanded focus to include diverse areas like art and entertainment, stressing the importance of cultural values in design (Benford et al., 2015) and first-person methods (Desjardins et al., 2021). Some examples from HCI and STS include the exploration of rural islandness (Robinson et al., 2021), the anthropological conceptualization of hope in the context of Sub-Saharan Africa (Ratto and Jackson, 2023), and the role of witchcraft in computing through villages of Bangladesh (Sultana and Ahmed, 2019).

However, it seems that cultural context often becomes a point of discussion only when research moves beyond urban, Western, and commonly studied contexts. In our own review, we focused exclusively on articles written in English. Partly because of this bias, research contexts are primarily based in Western countries, and the studies are typically conducted within researchers’ own local settings, while socio-geographic and cultural contexts are merely mentioned. However, it’s important to recognize that Western contexts are not universal—they are specific cultural environments in their own right. Expanding HCI to include cultural perspectives is valuable, but it doesn’t automatically address these complexities. Even within Western settings, when working in global organizations, understanding the local context remains important. This suggests there is room to further develop our awareness and sensitivity to cultural differences in research.

### 6.2.4. Considering entanglements of design ethics

Another point we want to bring into the discussion is the role of human and non-human entanglements in design ethics, building on Frauenberger’s explorations of STS and critical theories (Frauenberger, 2020). Verbeek (2006) explores the role of technological artifacts through the concept of technological mediation, highlighting how designed artifacts and society interact in ways that inherently involve

ethical considerations. This aligns well with the turn to practice in design ethics, where every interaction within assemblages of human and non-human actors is framed as an ethical encounter, and with practice theory that considers artifacts as crucial actors within practices (Kuijjer et al., 2013; Entwistle et al., 2015). This idea extends to the everyday tools and software design practitioners use, which influence their ethical decision-making processes.

Frauenberger (2020) argues that the concept of entanglement may represent the next major paradigm in HCI, as the field faces increasing pressure from the evolving relationship between humans and digital technologies. He observes that the rapid pace of technological innovation and profound social transformations are exposing the limitations of HCI’s existing foundations such as the inability to address ontological uncertainty, limitations in knowledge production, difficulties in assigning responsibility within complex socio-technical systems, reliance on a static concept of the user, and outdated ethical frameworks that fail to reflect the dynamic and relational nature of human-technology entanglements (Fuchsberger and C.F, 2025; Frauenberger, 2020). In the context of design ethics, the entanglement concept is suggested to imply a significant shift in how ethics (Frauenberger, 2020) and responsibilities (Fuchsberger and C.F, 2025) are understood—particularly as technologies gain autonomy and exert influence beyond direct human control. Fuchsberger and C.F (2025) highlight the resulting “responsibility gap,” where it becomes difficult to assign blame or accountability through traditional frameworks. They address this challenge by emphasizing that responsibility emerges through ongoing interactions among human and nonhuman actors across all stages of technological development. For design practice, this perspective means treating technology as an active participant, focusing on configuring relationships rather than isolated actions, and adopting inclusive, participatory methods that recognize the complex interplay of actors involved in shaping technological futures (Fuchsberger and C.F, 2025).

We observe that the subjects included in the studies we reviewed, predominantly practitioners, function as human actors within the design process. While our review includes research on tools and methods specifically designed to support ethical decision-making (e.g., Gray et al., 2022), there is a lack of studies addressing the interconnections between everyday design tools, software, and practitioners’ ethical decision-making. Entanglement theories driven by STS and critical theories can provide expansion to practice-oriented ethics research. Empirically, we suggest that this approach encourages us to foreground the materialities of the design process as research subjects. It highlights the active role these tools play as ethical mediators and actors within the process. Exploring this perspective further could inform the development of design tools that align more closely with the ethical and practical dimensions of the design process.

### 6.2.5. Developing specialized design methods for cultivation

This research documents the interventional methods employed in design ethics research. The interventional quality of these methods is distinctive to the way ethics is approached from a practice-oriented perspective. Within this perspective, ethics is not seen as a fixed set of principles but as something that is evolving, collaboratively constructed, and actively cultivated through practice. This framing of ethics calls for methodological approaches that go beyond traditional qualitative tools. Designery methods—particularly speculative and co-design approaches—are especially valuable in this context.

There is already an ongoing body of interventional methods developed by HCI and design researchers that contribute meaningfully to ethics research, particularly in fostering a collaborative ethos and cultivating shared ethical understanding. Speculative methods and design fiction, for instance, have long served as tools for critical reflection on emerging technologies. These methods make use of imaginative scenarios and designed artifacts to probe ethical tensions, stimulate discussion, engage diverse perspectives, and encourage critical reflection on the values that underpin technological development (Haghighi et al., 2023;

Wong and Nguyen, 2021; Tekogul and Forlano, 2024). Similarly, co-design and participatory design emphasize the inherently collaborative and situated nature of ethical inquiry. They provide interactive formats for exploring ethical concerns with stakeholders, reinforcing the view of ethics as cumulative, contextual, and socially negotiated (Lindberg et al., 2020, 2022). These approaches provide the conceptual and practical tools needed to support collaborative, reflective, and practice-based engagement with ethical challenges in design.

### 6.2.6. Making ethics engaging

A general observation we had is that the language surrounding ethical issues often carries negative connotations, such as avoiding deceptive designs, minimizing risks, and avoiding harmful future implications. Instead of framing ethics as the avoidance of unfavorable outcomes, how can we position it as something engaging and approachable? We find traces of this question in the review; for instance, Shilton (2013) positions design activities as levers that can “promote” social values, Lindberg et al. (2022) mentions the importance of “presenting ethics in an appealing manner”, and Luján Escalante et al. (2021) positions ethics as a capacity rather than a duty. This is an open question that we propose for future research to explore in a more structured way.

## 7. Conclusion

We have presented a characterization of practice-oriented research in design ethics in relation to the turn to practice in HCI research. The paper highlights a growing interest in understanding and addressing the ethical implications of design, emphasizing the value of practice-oriented approaches in capturing the situated, tacit, and complex nature of design ethics. It also demonstrates the potential of such approaches to develop concepts, tools, and methods that better align with the realities of design practice.

The paper’s findings highlight several future directions that need to be addressed to advance research on design ethics in practice. While empirical studies in this area are increasing, there remains a significant lack of concept-building empirical work. Current research predominantly relies on cross-sectional study designs with interviews as the primary method, highlighting the need for long-term studies to enhance understanding and address the temporality of ethics. Design ethics are recognized as socially and culturally situated, but further research is needed to socio-geographically contextualize these studies. Although existing research explores specialized tools and methods for ethical consideration in the design process, there is a need to examine how existing design tools mediate ethical practices. The paper also emphasizes the significance of design methods that can bring collaborative approaches to cultivation and position ethics as an active, engaging, and resourceful capacity in design, rather than approaching it with caution or hesitation. Finally, we note that design ethics research spans a broad spectrum of fields, with the design in HCI research well-positioned to guide design ethics in practice studies across disciplines, with its cumulative experience in practice-oriented research for grappling with various impacts of technology development.

### CRedit authorship contribution statement

**Gizem Öz:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Christian Dindler:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Sharon Lindberg:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization.

### Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

I have nothing to declare. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### Data availability

No data was used for the research described in the article.

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